

The symmetric convex body of contraction minors

MathlibAnnex.PluckerBody.body

def

Takes the real convex hull of both signs of every contraction generator.

Statement

Let $n, N \in \mathbb{N}_0$ and give $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ their coordinate sup norms. Let λ_n be the fixed coordinate Lebesgue measure on E . Let $p : E \rightarrow \mathbb{R}$ be a continuous real seminorm with specified constants $a_p, b_p > 0$ such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in E).$$

The lower bound makes p a norm. Its closed unit ball and real reference volume are

$$K_p = \{x : p(x) \leq 1\}, \quad v_p = \lambda_n(K_p) \in (0, \infty).$$

Here and below a finite Lebesgue measure is read as a real number when used as a scalar.

Let $\mathcal{I}_{n,N}$ be the finite set of increasing n -tuples of distinct rows from $\{1, \dots, N\}$. The minor-coordinate space $\mathbb{R}^{\mathcal{I}_{n,N}}$ carries its sup norm. For a linear map $B : E \rightarrow F$, put $B_{ij} = (Be_j)_i$ in the standard coordinate basis. Write $\Pi_n(B)$ for its maximal-minor vector and $\mathcal{V}_p(B)$ for that vector scaled by v_p . Their coordinates and scaling are

$$\Pi_n(B)_I = \det(B_{i_a,j})_{a,j=1}^n, \quad \mathcal{V}_p(B) = v_p \Pi_n(B) \quad (I = (i_1 < \dots < i_n)).$$

A p -contraction is a linear map $B : E \rightarrow F$ satisfying $\|Bz\|_\infty \leq p(z)$ for every $z \in E$. Write $\mathcal{C}_p^N$ for these maps. Write $\mathcal{G}_p^N$ for the signed generator set and $\mathcal{P}_p^N$ for its real convex hull, so

$$\mathcal{G}_p^N = \{\mathcal{V}_p(B), -\mathcal{V}_p(B) : B \in \mathcal{C}_p^N\}, \quad \mathcal{P}_p^N = \text{conv}_{\mathbb{R}}(\mathcal{G}_p^N).$$

Definition

For each integer $\ell \geq 1$, choose weights $t_1, \dots, t_\ell \geq 0$ with sum 1, signs $\varepsilon_k \in \{-1, 1\}$, and maps $B_k \in \mathcal{C}_p^N$. The body consists of all resulting finite convex combinations:

$$\mathcal{P}_p^N = \left\{ \sum_{k=1}^{\ell} t_k \varepsilon_k \mathcal{V}_p(B_k) : \ell \geq 1, t_k \geq 0, \sum_{k=1}^{\ell} t_k = 1, \varepsilon_k \in \{-1, 1\}, B_k \in \mathcal{C}_p^N \right\}.$$

The signed-generator definition is a separate declaration. The exact RHS of the present definition takes its real convex hull.

Assumptions

The source and target are finite real coordinate spaces with sup norms. The seminorm p is continuous and has the stated positive lower and upper comparison constants. The measure is the original coordinate Lebesgue measure. The dimensions n, N may be zero. When $n = 0$, the unique empty minor has determinant 1; when $N < n$, the minor-index set is empty and its function space has one element. Neither case is excluded.

Conclusion

This declaration defines $\mathcal{P}_p^N$. Compactness and average membership are separate theorems.

The zero map is a contraction, so generator nonemptiness supplies $\mathcal{V}_p(0) \in \mathcal{G}_p^N$; this vector need not be zero when $n = 0$. The two alternatives give generator sign symmetry. Consequently body sign symmetry says $z \in \mathcal{P}_p^N \Rightarrow -z \in \mathcal{P}_p^N$: negate each generator in a convex combination while keeping its nonnegative weights.

Main citations

- Exact declaration and its source — `MathlibAnnex.PluckerBody.body`
- signed-generator definition — `MathlibAnnex.PluckerBody.generators`
- generator nonemptiness — `MathlibAnnex.PluckerBody.generators_nonempty`
- generator sign symmetry — `MathlibAnnex.PluckerBody.generators_neg`
- body sign symmetry — `MathlibAnnex.PluckerBody.body_neg`

Lean source signature (exact)

```
/-- The convex hull of all signed model-contraction generators. -/
def body {n : ℕ} (M : EquivalentSeminorm (Fin n → ℝ)) (N : ℕ) :
  Set (Matrix.MaximalMinorIndex n (Fin N) → ℝ) := convexHull ℝ (generators M N)
```

This is a definition of a set. The final `:=` gives its defining expression.

In the source	Mathematical meaning
<code>{n : ℕ}; (N : ℕ)</code>	Nonnegative integers n, N . Lean may infer n from <code>M</code> ; the output dimension N is an explicit argument of <code>body</code> .
<code>Fin n → ℝ; Fin N → ℝ</code>	The spaces $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ with their sup norms. A vector is a list of real coordinates, indexed from 0 in Lean and from 1 in the formulas here.
<code>M : EquivalentSeminorm (Fin n → ℝ)</code>	The continuous seminorm $p = M.p$, with $a_p = M.lower > 0$, $b_p = M.upper > 0$ and $a_p \ x\ _\infty \leq p(x) \leq b_p \ x\ _\infty$ for every $x \in E$.
<code>Matrix.MaximalMinorIndex n (Fin N)</code>	The index set $\mathcal{I}_{n,N}$: choose n distinct rows among the N rows and read them in increasing order. A function of this type into $\mathbb{R}$ is a minor-coordinate vector $z = (z_I)_{I \in \mathcal{I}_{n,N}}$.

In the source	Mathematical meaning
<code>M.closedUnitBall; M.closedUnitBallVolume</code>	$K_p = \{x : p(x) \leq 1\}$ and $v_p = \lambda_n(K_p) > 0$, with the finite measure read as a real number.
<code>M.IsContraction B</code>	$\ Bz\ _\infty \leq p(z)$ for every $z \in E$, where $B : E \rightarrow F$ is real-linear.
<code>Matrix.ballVolumeScaledMaximalMinors M B</code>	$\mathcal{V}_p(B) = v_p \Pi_n(B)$, whose $I = (i_1 < \dots < i_n)$ coordinate is $v_p \det \left((Be_j)_{i_a} \right)_{a,j=1}^n$.
<code>generators M N</code>	The set $\mathcal{G}_p^N = \{\pm v_p \Pi_n(B) : B : E \rightarrow F \text{ linear}, \forall z \in E, \ Bz\ _\infty \leq p(z)\}$. Its elements are minor-coordinate vectors, not the maps B .

In the source	Mathematical meaning
<code>convexHull ℝ (generators M N)</code>	The real convex hull of that set: $\mathcal{P}_p^N$. A point is a finite sum $\sum_k t_k z_k$ with $t_k \geq 0$, $\sum_k t_k = 1$, and $z_k \in \mathcal{G}_p^N$.
<code>Set (Matrix.MaximalMinorIndex n (Fin N) → ℝ)</code>	A set of minor-coordinate vectors, so the output is a subset of $\mathbb{R}^{\mathcal{I}_{n,N}}$.

`generators M N` applies the definition `generators` to `M` and `N`; it does not multiply them.

Exact definitions used in this reading: Signed generator set.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 12–14:

```

noncomputable section

namespace MathlibAnnex.PluckerBody

```

▼ Exact content identity

Declaration: `MathlibAnnex.PluckerBody.body`
Accepted content SHA-256: 47aa096afcfbabba850a581df4da7a16644022421b88caf15f80f6542bb10efd
Accepted source guide SHA-256: 1d77d3917585faa78f7a5b0bf31a2688b36eb63227f06925b19410275f2f65f3
Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73